Hospital Value-Performance Benchmark

Quality vs. Cost Opportunity Finder

Project Documentation: Assumptions, Decisions, and Findings

1. Overview

This document records every assumption, judgment call, bug found and fixed, and key finding made while building the Hospital Value-Performance Benchmark pipeline. The source data is CMS's public Provider Data Catalog hospital datasets. The pipeline joins quality, readmission, patient experience (HCAHPS), and Medicare Spending Per Beneficiary (MSPB) data; builds a composite value score; groups hospitals into peer groups; detects outliers; trains a baseline machine learning model; and explains individual hospital scores using SHAP values, all surfaced through an interactive Streamlit dashboard.

2. Data Sources Used

Out of 74 available CMS files, four were selected as the minimum set needed to answer the core question of quality vs. cost:

- Hospital_General_Information.csv — overall star rating, hospital type, ownership, emergency services (used for quality scoring and peer grouping)
- HCAHPS-Hospital.csv — patient experience survey results (quality)
- Unplanned_Hospital_Visits-Hospital.csv — 30-day readmission rates for six conditions (quality)
- HOSPITAL_QUARTERLY_MSPB_6_DECIMALS.csv — Medicare Spending Per Beneficiary (cost)

Assumption: the remaining 70 files (infection rates, complications, imaging efficiency, state/national summary files, psychiatric-specific and veterans' hospital files, etc.) were excluded as out of scope for this specific quality-vs-cost question. This is a documented limitation, not an oversight — see Section 8.

3. Stage 2 — Cleaning & Joining

Bug found and fixed: H_STAR_RATING column 100% missing.

The initial pivot of HCAHPS-Hospital.csv read values only from the "HCAHPS Answer Percent" column. CMS actually splits values across three different columns depending on measure type: "HCAHPS Answer Percent" (percentage measures), "HCAHPS Linear Mean Value" (0-100 score measures), and "Patient Survey Star Rating" (star measures). Since H_STAR_RATING is a star measure, its value was never being read, producing 100% missing data.

Fix: all three columns were coalesced into a single unified value column (using combine_first logic) before pivoting, since each row only ever populates exactly one of the three columns. After the fix, H_STAR_RATING missingness dropped from 100% to approximately 41.5%, consistent with the other quality columns.

Assumption: CMS's text placeholders ("Not Applicable", "Not Available", "N/A", blank) are all treated as missing (NaN) uniformly across every file.

4. Stage 3 — Feature Engineering

Core features built:

- quality_overall — CMS overall star rating (1–5, higher = better)
- quality_patient_exp — H_STAR_RATING patient experience rating (1–5, higher = better)
- readmission_rate_avg — average of six 30-day readmission rates (AMI, CABG, COPD, HF, HIP_KNEE, PN); lower = better
- cost_value — MSPB Medicare spending per beneficiary; lower = better

Documented assumptions:

- Missing values are left as NaN and skipped per-hospital, per-calculation, rather than filled with dataset-wide averages. This avoids blurring the real signal that "missing" often means "too small to report" rather than "bad."
- All six readmission measures are weighted equally; no adjustment for how common or rare each condition is.
- Quality and cost are combined with equal weight (50% / 50%) in the composite value_score. This is a starting assumption, adjustable if a different priority between quality and cost is desired.
- Z-scores (standardized scores) are used so percentages, dollar amounts, and star ratings can be compared on the same scale. Readmission rate and cost z-scores are sign-flipped so that, consistently across every metric, a positive z-score always means "better."

5. Stage 4 — Peer Grouping

Hospitals were grouped using K-means clustering on three characteristics: Hospital Type, Hospital Ownership, and Emergency Services. The number of groups (k = 4) was chosen using the elbow method, testing k = 2 through 10 and selecting the point where additional groups stopped meaningfully improving cluster tightness.

Documented limitation: no bed-size or teaching-hospital status data.

CMS's Provider Data Catalog does not include hospital bed count or academic/teaching status; this data resides in a separate system (Medicare Cost Reports / HCRIS) not part of this file set. Peer grouping therefore relies on hospital type, ownership, and emergency services only. This is a known, documented limitation; adding bed-size data as a clustering feature is a recommended future improvement.

Decision: State excluded from clustering math.

Including State as a clustering feature would cause the algorithm to treat "different state" as a large distance even between otherwise identical hospitals, effectively recreating 50 state-based clusters instead of finding operationally similar hospitals. State is instead retained as an independent dashboard filter, layered on top of the peer group filter.

Feature dropped: birthing-friendly designation.

“Meets criteria for birthing friendly designation” was initially included as a fourth clustering feature but was found to artificially split otherwise identical hospitals into duplicate clusters, based only on whether a hospital had applied for this specific voluntary maternity-care certification (mostly blank, not a meaningful “No”). This feature was removed from clustering after manual verification of the duplicate clusters.

Result: four peer groups — (1) Acute Care Hospitals / Voluntary non-profit — Private (2,333 hospitals), (2) Critical Access Hospitals / Voluntary non-profit — Private (1,609 hospitals), (3) Psychiatric / Proprietary (798 hospitals), (4) Acute Care Hospitals / Proprietary (692 hospitals).

6. Stage 5 — Outlier Detection

Within each peer group, hospitals are classified using their peer-group-relative value_score_peer: Strong positive outlier (≥ +2.0 standard deviations), Strong negative outlier (≤ −2.0 standard deviations), Typical for peer group (between −2.0 and +2.0), or Insufficient data (no usable metrics). The 2.0 standard deviation threshold is a standard statistical convention, corresponding to roughly the top and bottom 5% of a normal distribution, and is adjustable via a single variable in the code.

Bug found and fixed: composite score formula silently discarded valid quality data.

The original value_score_peer formula computed 0.5 × quality_component + 0.5 × cost_component directly. Because arithmetic involving a missing value (NaN) in Python/pandas returns NaN, any hospital missing its cost component received a fully blank score — even when its quality data was complete. This affected all 1,609 Critical Access Hospitals, since CMS does not calculate MSPB spending for this hospital type, causing 100% of this peer group to show “Insufficient data” despite roughly 57% of them having valid readmission data.

Fix: the formula was replaced with a weighted average that renormalizes weights across only the available components, rather than failing when one component is missing. After the fix, hospitals with a usable score increased from 2,819 to 3,970 (out of 5,432, approximately 73%).

Result: 61 hospitals nationally flagged as strong positive outliers; 24 flagged as strong negative outliers, each with an identified single biggest driver (“biggest_drag_metric”) and biggest strength.

7. Stage 6 — Baseline Model

A Random Forest / Gradient Boosting regression model was trained to predict CMS's official “Hospital overall rating” from engineered features, rather than relying solely on the hand-built value_score formula. Hospitals missing the target rating were excluded from training; missing inputs were filled using peer-group averages specifically for model training (the original feature files retain NaNs untouched).

Model iteration and results:

Version	Model / Features	MAE (stars)	R^2	Notes
v1	Random Forest, 4 features	0.807	0.157	Baseline

v2	Random Forest, 7 features (+MORT/Safety/READM net)	0.662	0.422	Added CMS category data
v3	Gradient Boosting, same 7 features	0.603	0.512	Final baseline model

Assumption: quality_overall (the prediction target) is deliberately excluded from the model's input features to prevent the model from simply copying the answer rather than learning a genuine relationship.

Documented limitation: R^2 of 0.512 means the model explains just over half of what determines a hospital's official CMS rating. This is expected and reasonable, since CMS's real methodology blends additional measure categories (e.g., Timeliness, further Patient Experience sub-measures, infection rates) not included in this pipeline's scope. A Timeliness “net” feature analogous to MORT_net/Safety_net/READM_net was considered but found not to be constructible, since CMS does not provide a Better/Worse breakdown for the Timeliness or Patient Experience measure groups, only a count of measures reported.

Finding: readmission rate and mortality performance were consistently the strongest predictors of official rating across model versions; patient experience became the top predictor once Gradient Boosting (v3) was used.

8. Stage 7 — Explainability (SHAP)

SHAP (SHapley Additive exPlanations) values were calculated for every hospital using the final v3 Gradient Boosting model, providing a hospital-specific breakdown of how much each feature added to or subtracted from that hospital's predicted rating, relative to a baseline average prediction.

Documented limitation: imputed inputs produce less meaningful explanations.

Because missing model inputs were filled with peer-group averages prior to prediction, a hospital missing a given raw metric (e.g., cost) will show a SHAP contribution for that feature reflecting the peer-group average rather than a genuine hospital-specific value. This is a necessary tradeoff to allow scoring of hospitals with partial data, and is noted here so imputed-feature explanations are not mistaken for hospital-specific findings.

Finding: patient experience is simultaneously the most common top positive driver (2,065 hospitals) and the most common top negative driver (1,883 hospitals) nationally — it is the single most influential factor overall, capable of swinging scores strongly in either direction. Safety performance (Safety_net) is disproportionately a weakness rather than a strength across hospitals (1,814 hospitals vs. 627), suggesting broad room for improvement in this category specifically.

9. Stage 8 — Dashboard

An interactive Streamlit dashboard was built using Plotly for visualizations. Users can filter by State and Peer Group, select an individual hospital, and view: its peer-relative value score and outlier status; a SHAP-based bar chart showing which specific metrics helped or hurt its predicted rating; a radar chart comparing the hospital to its peer group average across four key metrics; and a table of the strongest positive and negative outliers within the current filter selection.

10. Summary of All Documented Limitations

- No bed-size or teaching-hospital-status data available in this file set; peer grouping is based on hospital type, ownership, and emergency services only.
- Approximately 27% of hospitals (1,462 of 5,432) have insufficient data across all four core metrics and cannot be scored under this methodology, primarily Psychiatric hospitals and a subset of Critical Access Hospitals.
- The baseline model explains approximately 51% ($R^2 = 0.512$) of the variation in official CMS ratings; the remainder reflects measure categories outside this pipeline's scope (e.g., Timeliness, infection rates, complications).
- SHAP explanations for hospitals with imputed (peer-group-averaged) inputs reflect the peer group average for that feature, not a hospital-specific value.
- 70 of 74 available CMS files were excluded as out of scope for the quality-vs-cost question; incorporating additional files (e.g., Healthcare_Associated_Infections-Hospital.csv, Complications_and_Deaths-Hospital.csv) is a documented direction for future improvement.